



L&T Technology Solutions works with companies to set up digital twins or virtual replicas of their factories

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between a hardhat and a regular cap. Or it would reject hats of a different colour. The system took time to learn but that's part of the process," he says.

Tata Steel's Bannerjee sounds caution. "Fundamentally, AI is not a standalone journey. You need to first invest in Cloud, Edge and data technology. You need to build the road before the car. That has been our transformation journey at Tata Steel. We've invested in the Cloud and are working on integrating the data layer at all our sites in India and worldwide."

LTTS' Chadha also emphasises the importance of having robust data management practices to benefit from AI. He cites the example of a European industrial manufacturer. "The biggest hurdle was not AI algorithms but unifying disparate data sources from decades-old systems. Once harmonised, AI-driven twins unlocked predictive maintenance insights that reduced unplanned downtime by 15%," he says.

For Indian enterprises, this is turning out to be among the biggest hurdles to implementing new technologies. Apart from the larger conglomerates, a majority of manufacturing facilities still use old machinery which doesn't have IoT capabilities. Investing in newer machines and collecting and collating this data is the first step towards an AI integration road map.

Trust is another factor. AI adoption is a multi-pronged change management exercise where building trust among engineers and operators in the AI's recommendations is as important as building a robust and reliable AI algorithm.

BCG's Ghosh says once there is a general acceptance that AI can deliver results, the second hurdle is achieving economies of scale. "It becomes a value versus cost play. If there is a proven value, as against what you must pay, there will be adoption," she says. Given that the technology is still at a nascent stage, the cost of the AI integration may be more than the productivity gains in some cases.

The next stage is likely to see the use of generative AI-based agents on the shop floor. Global engineering major Siemens has been investing in integrating AI tools into its entire range of offerings. "Industrial AI is the key to transforming industries. Generative AI will be a valuable tool for every engineer and an indispensable part of the future engineering process across all industries," says a Siemens India spokesperson. This will tackle critical industrial challenges like high machine downtime, time intense failure diagnosis and response.

Companies must realise that deciding whether to bring physical AI tools on their shop floor is no longer a question of if but when. There is a shift from batch and mass production towards customisation. Manufacturing will have to evolve to meet these changing customer demands, and AI powered automation will be the most effective way to get there. **BT**